# Anderson Localization in Random Band Matrices and Its Application to Long-Range Resonator Models

A Report for

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**Abstract**

In this report, we investigated Anderson localisation in random band matrices and its applications to one-dimensional chains of acoustic resonators. In particular, we were motivated by the work of Ammari et al. [3][4][2][5], who studied localisation in disordered acoustic resonator systems. More recently, Ammari et al.[5] demonstrated that the interaction coefficients of a one-dimensional resonator chain embedded in three-dimensional space exhibit long-range decay of the form

$$|c_r| \asymp \frac{1}{|r|(\log|r|)^2}.$$

This motivated our investigation into whether fractional-moment methods, as developed by Schenker [8], could be applied to such resonator models.

# 1 What is localisation?

In this section, we introduce the concept of *eigenvector localisation*.

## 1.1 Anderson Localisation

In 1958, P. W. Anderson studied the propagation of electrons in disordered lattices [6]. He considered a model in which electrons may move between neighbouring lattice sites, while the energy associated with each site is random.

In one dimension, a finite-dimensional version of this model is represented by the random matrix

$$H_N = \begin{pmatrix} v_1 & t & 0 & \cdots & 0 \\ t & v_2 & t & \ddots & \vdots \\ 0 & t & v_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & t \\ 0 & \cdots & 0 & t & v_N \end{pmatrix} \tag{1}$$

where $t \neq 0$ represents the hopping amplitude between neighbouring sites, and $v_1, \ldots, v_N$ are independent random variables representing the energies at individual sites.

When the site energies are identical, the eigenvectors of $H_N$ are extended across the lattice. Anderson showed that disorder can instead suppress the diffusion of electrons, even though hopping between sites remains possible. In the localised regime, the eigenstates are concentrated around particular sites rather than extending throughout the lattice.

This phenomenon, now known as *Anderson localisation*, motivates the study of eigenvector localisation in more general random matrix models.

## 1.2 Localisation of Band Matrices

Eigenvector localisation is now studied for more general random matrix models. In particular, we are interested in *random band matrices* with a width denoted as $W$.

**Definition 1.1** (Random Band Matrix)**.** Let $N, W \in \mathbb{N}$, where $1 \leq W \leq N$. An $N \times N$ random Hermitian matrix

$$H_{W;N} = (h_{ij})_{i,j=1}^{N}$$

is called a *random band matrix* of bandwidth $W$ if

$$h_{ij} = 0 \qquad \text{whenever } |i-j| \geq W.$$

More explicitly, its entries may be written as

$$h_{ij} = \begin{cases} v_i, & i = j, \\ t_{ij}, & 0 < j - i < W, \\ \overline{t_{ji}}, & 0 < i - j < W, \\ 0, & |i-j| \geq W, \end{cases}$$

where $v_i \in \mathbb{R}$ and $t_{ij} \in \mathbb{C}$ are random variables.

An important question in the study of random band matrices is how the localisation properties of their eigenvectors depend on the bandwidth $W$ as the matrix dimension $N$ tends to infinity. For standard one-dimensional random band matrix ensembles, the characteristic localisation length is predicted to be of order $W^2$.

In 2009, [8] established localisation for certain random band matrix ensembles by proving exponential decay estimates for fractional moments of their Green's functions. For Gaussian band matrices, his result gives a localisation length bound of order $W^8$, and consequently establishes localisation in the regime

$$W^8 \ll N.$$

He showed this by showing that the fractional moment of the Green function decays exponentially. We expound on this more in the next subsection.

### 1.3 Schenker's Fractional Moment Analysis

To understand Schenker's approach, we first introduce the *Green's function* associated with a random band matrix.

**Definition 1.2** (Green Function)**.** Let $H_{W;N}$ be an $N \times N$ random Hermitian band matrix. For $\lambda \notin \sigma(H_{W;N})$, we define its Green function by

$$G_{W;N}(i, j; \lambda) = \left\langle e_i, (H_{W;N} - \lambda I)^{-1} e_j \right\rangle$$

where $e_i$ denotes the $i$th standard basis vector. The matrix $(H_{W;N} - zI)^{-1}$ is called the *resolvent* of $H_{W;N}$.

The reason why localisation of eigenvectors are related to the Green function can be shown via the following formula: Let $\lambda_k$ and $v_k$ denote its eigenvalues and corresponding orthonormal eigenvectors, then by the spectral theorem,

$$G_{W;N}(i, j; \lambda) = \sum_{k=1}^{N} \frac{v_k(i)\overline{v_k(j)}}{\lambda_k - \lambda}$$

Thus, the Green's function contains information about the overlap of eigenvector amplitudes at two sites $i$ and $j$. In particular, localisation suggests that these overlaps should become small as the distance $|i-j|$ increases.

Instead of studying the eigenvectors directly, Schenker [8] investigates the *fractional moments* of the Green's function, namely

$$\mathbb{E}\left(|G_{W;N}(i,j;\lambda)|^s\right) \qquad 0<s<1$$

The use of fractional powers is important because the resolvent can become arbitrarily large when $\lambda$ approaches an eigenvalue. Fractional powers reduce the contribution of such large values.

His goal was to establish an estimate of the form

$$\mathbb{E}\left(|G_{W;N}(i,j;\lambda)|^s\right) \leq A_s W^{s\sigma} \exp\left(-\alpha_s \frac{|i-j|}{W^\mu}\right)$$

This states that the fractional moments of the Green's function decay exponentially with the distance between two sites.

Schenker establishes this estimate using two complementary properties.

**Lemma W (Weak-tail estimate). [8]** The probability that the Green's function is exceptionally large is bounded by

$$\mathbb{P}\left(|G_{W;N}(i,j;\lambda)|>t\right) \leq \frac{\kappa W^\sigma}{t}$$

In particular, this implies that its fractional moments are finite and uniformly bounded in the matrix dimension $N$.

**Lemma F (Fluctuation estimate). [8]** For $0<r<s<1$, Schenker establishes an improvement to Hölder's inequality of the form

$$\mathbb{E}\left(|G_{W;N}(i,j;\lambda)|^r\right) \leq \exp\left(-c_{r,s}\frac{|i-j|}{W^\mu}\right)\mathbb{E}\left(|G_{W;N}(i,j;\lambda)|^s\right)^{r/s}$$

for sufficiently separated sites.

The exponential factor arises from fluctuations of the logarithm of the Green's function. Intuitively, the accumulated fluctuations between distant sites produce an increasingly strong improvement to Hölder's inequality.

Combining Lemmas W and F yields

$$\mathbb{E}|G_{W;N}(i,j;\lambda)|^r \leq C_r W^{\sigma r} e^{-c_{r,s}|i-j|/W^\mu}$$

Finally, using additional spectral-averaging arguments, this fractional-moment decay can be converted into an eigenvector localisation estimate. More precisely, Schenker's Theorem 2 establishes, under the relevant assumptions, that

$$\mathbb{E}\left[\sup_{\lambda_k\in[-R,R]}|v_k(i)v_k(j)|\right] \leq BW^\tau e^{-\beta|i-j|/W^\mu}$$

This shows that eigenvectors have exponentially suppressed overlap between sufficiently distant sites. For the Gaussian band ensemble, Schenker obtains $\mu=8$.

Hence, the fractional-moment method provides a way to establish eigenvector localisation through the decay properties of the Green's function, without directly analysing the eigenvectors themselves.

More recently, [7] proved exponential eigenvector localisation on the sharp scale $W^2$ for a broad class of one-dimensional random band matrices, under the condition

$$W^2 \ll N$$

### 1.4 Experiment 1: Localisation of Band Matrices

In this subsection, we look at how eigenvector localisation should look like in terms of a spectra graph.

- **Matrix size and bandwidth:** $N = 500$, $W = 5$, so that

$$\frac{W^2}{N} = \frac{25}{500} = 0.05 \ll 1$$

- **Matrix ensemble:** Complex Hermitian Gaussian random band matrix $H \in \mathbb{C}^{N\times N}$, satisfying $H_{ji} = \overline{H_{ij}}$.
  - **Diagonal entries:**

$$H_{ii} \sim \mathcal{N}\left(0, \frac{1}{4W}\right).$$

  - **Off-diagonal entries:** For $i < j$ with $|i - j| < W$, let

$$H_{ij} = X_{ij} + iY_{ij}$$

    where

$$X_{ij}, Y_{ij} \overset{\text{ind}}{\sim} \mathcal{N}\left(0, \frac{1}{8W}\right)$$

- **Randomness:** All diagonal entries and upper-triangular Gaussian components are mutually independent. The lower-triangular entries are determined by Hermitian symmetry.

Figure 1 shows the eigenvector component distribution of the $k = 184$-th eigenvector. We can see clearly that a good proportion of the components are at (or close to) 0 except for a region, with a clear peak. This phenomenon is localisation.

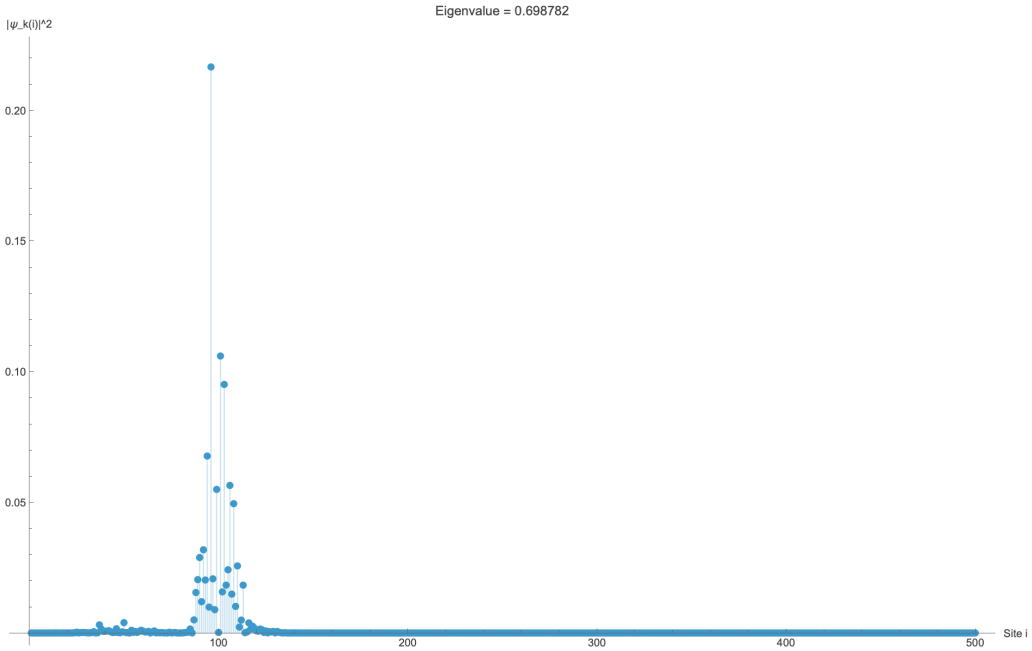



Figure 1: Spatial profile of the eigenvector corresponding to $\lambda_{k=184} \approx 0.699$ for a Gaussian random band matrix with $N = 500$ and $W = 5$.

# 2 Localisation in Long-range Resonator Chains

## 2.1 Motivations

In the study of Anderson localisation, one of the models that are relevant are acoustic resonators. Acoustic resonators are essentially objects that naturally oscillate at certain frequencies of sound.

To understand why eigenvectors arise naturally in this setting, we first consider the equations governing the vibrations of coupled oscillators. Let $u_i(t)$ denote the displacement of the $i$-th oscillator from its equilibrium position at time $t$, and write

$$u(t) = \begin{pmatrix} u_1(t) & \dots & u_N(t) \end{pmatrix}$$

In a simplified linear model without damping or external forcing, the equations of motion take the form

$$M\ddot{u}(t) + Ku(t) = 0$$

where $M$ is a positive definite mass matrix and $K$ is a symmetric stiffness matrix describing the restoring forces and interactions between the oscillators.

We are particularly interested in the natural modes of vibration of the system. These are oscillations in which all oscillators vibrate at the same frequency, although their amplitudes may differ. To find such modes, we seek solutions of the form

$$u(t) = \phi e^{i\omega t}$$

where $\omega$ is the angular frequency and $\phi$ is a vector describing the relative amplitudes of the oscillators.

Differentiating twice gives

$$\ddot{u}(t) = -\omega^2 \phi e^{i\omega t}$$

Substituting into the equation of motion, we obtain

$$-\omega^2 M\phi e^{i\omega t} + K\phi e^{i\omega t} = 0$$

Since $e^{i\omega t} \neq 0$, we get

$$K\phi = \omega^2 M\phi$$

Thus, finding the natural vibrational modes of the coupled system amounts to solving a generalised eigenvalue problem. In particular, by defining

$$H = M^{-1/2} K M^{-1/2} \qquad \psi = M^{1/2}\phi$$

we obtain

$$H\psi = \omega^2 \psi$$

This explains the connection between eigenvectors and vibrations. The eigenvalues determine the squared natural frequencies, while the corresponding eigenvectors determine the relative oscillation amplitudes of the resonators, after accounting for the change of coordinates.

For example, if a mode has the amplitude distribution

$$\phi = \begin{pmatrix} 0.1 & 0.2 & 0.8 & 0.1 & 0.2 \end{pmatrix}$$

then the third oscillator vibrates with a much larger amplitude than the others in that particular mode.

The capacitance matrix encodes interactions between resonators, and its appropriately weighted eigenvectors describe the spatial profiles of the resonant modes. This leads us naturally to the study of eigenvector localisation.

In a periodic chain of identical resonators, vibrational modes may extend throughout the system. However, when randomness is introduced into the resonators' material properties, the resulting eigenvectors may become concentrated around particular sites.

More precisely, if an eigenvector is exponentially localised around some site $i_0$, then its components satisfy a bound of the form

$$|\psi(i)| \leq Ce^{-\alpha|i-i_0|}$$

for some constants $C, \alpha > 0$.

Consequently, the oscillation amplitudes of resonators far from $i_0$ become exponentially small, provided the coordinate rescaling between the eigenvector and the physical mode is uniformly bounded. Physically, this means that the corresponding vibrational mode is effectively confined to a small region of the resonator chain, rather than extending throughout the entire system.

Therefore, studying localisation of the eigenvectors of the resonator matrix allows us to investigate the spatial confinement of acoustic resonant modes.

An important distinction: This phenomenon is different and distinct from energy dissipation of vibrations over time. In a physical resonator, vibrations may gradually weaken because energy is transferred to the surrounding air or converted into heat. Anderson localisation, however, can occur even in the absence of energy dissipation. Rather than describing how vibrations decay over time, it describes how the oscillation amplitudes of a particular resonant mode decay with distance from a localisation centre. In other words, the vibrational mode becomes spatially confined not because energy is lost, but because disorder causes repeated wave scattering and interference, suppressing its propagation through the resonator chain.

This motivates our central question: under what conditions can random variations in the material properties of coupled acoustic resonators produce localised vibrational modes?

The study of Anderson localisation in acoustic resonator systems has received considerable attention in recent years. More specifically, Ammari et al. [3] investigated localisation in systems of subwavelength resonators with random material parameters. Using generalised capacitance matrices, they developed a mathematical framework for studying the interactions between resonators and the resulting localised modes. Subsequently, Ammari et al. [4] extended this investigation to finite resonator systems with disorder in their spatial arrangements, studying various mechanisms of wave localisation.

More recently, Ammari et al. [5] considered a one-dimensional periodic chain of acoustic resonators embedded in three-dimensional space. They established that the interactions between distant resonators exhibit a long-range decay of the form

$$|c_r| \asymp \frac{1}{|r|(\log|r|)^2} \tag{2}$$

where $r$ denotes the separation between resonators. Although their numerical experiments suggest Anderson localisation in the presence of disorder, a rigorous proof for arbitrary nonzero disorder remains an open problem.

Motivated by these results, we investigate finite-band approximations of the random capacitance operator associated with acoustic resonator chains. In particular, we study the error introduced by truncating long-range interactions to a prescribed bandwidth $W$, before investigating localisation properties of the resulting random band matrices using fractional-moment methods.

## 2.2 Mathematical Formulation of the Resonator Model

Following the capacitance-matrix framework developed by Ammari et al. [3] and subsequently studied by Ammari et al. [5], we introduce a mathematical model describing the interactions between acoustic resonators arranged in a one-dimensional chain.

In particular, we represent these interactions using a capacitance matrix $C$, whose entries encode the coupling between individual resonators. Given an *infinite* chain of resonators, we define a *capacitance matrix* to be the following:

**Definition 2.1** (Capacitance Matrix)**.** A capacitance matrix $C$ is a matrix where:

1. The entries $c_{ij}$, where $c_{ij}$ represents the interaction coefficient between resonators $i$ and $j$. The interaction coefficient $c_{ii}$ will represent the self-interaction of resonator $i$.

2. $C$ is *Toeplitz*, that is, $C_{ij} = c_{i-j}$. This is because sites $i$ and $i+1$ should interact identically to $i+1$ and $i+2$. So this is equivalent to saying that $C$ is Toeplitz.

**Remark 2.2.** Note that $C$ being Toeplitz is only for an infinite-chain model. In a finite-chain, the first resonator is an edge, so it may not have the same surroundings as say the third resonator. And by construction, $C$ is guaranteed to be symmetric.

Now as explained, in our model we are formulating, each resonator have different properties. So, we model this by introducing positive, random parameters $\gamma_i > 0$ and define the matrix

$$V = \operatorname{diag}(\gamma_i)_{i\in\mathbb{Z}}$$

Then, we define the matrix $H$ to be

$$H = V^{1/2} C V^{1/2} = \left(\sqrt{\gamma_i \gamma_j}\, c_{i-j}\right)_{i,j\in\mathbb{Z}}$$

Its matrix representation is:

$$H = \begin{pmatrix} \ddots & \vdots & \vdots & \vdots & \vdots \\ \cdots & \gamma_{-1}c_0 & \sqrt{\gamma_{-1}\gamma_0}c_1 & \sqrt{\gamma_{-1}\gamma_1}c_2 & \cdots \\ \cdots & \sqrt{\gamma_0\gamma_{-1}}c_1 & \gamma_0 c_0 & \sqrt{\gamma_0\gamma_1}c_1 & \cdots \\ \cdots & \sqrt{\gamma_1\gamma_{-1}}c_2 & \sqrt{\gamma_1\gamma_0}c_1 & \gamma_1 c_0 & \cdots \\ \cdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

**Remark 2.3.** A natural representation of $H$ would be $VC$. However, that would make $H$ non-symmetric. Instead, the following construction gives us the same eigenvalues, while having $H$ to be Hermitian, which makes it suitable for our localisation analysis as shown in Chapter 1.

Another note is that $H$ is not necessarily Toeplitz, because the parameters $\gamma_i$ varies between sites. Therefore, the symmetry between parallel sites may no longer hold.

We now state the properties of $C$, as presented in [5].

First, we have the following theorem.,

**Theorem 2.4** ([5], Theorem 3.8)**.** *Let $C$ be an infinite capacitance matrix associated with a one-dimensional periodic chain of identical resonators embedded in $\mathbb{R}^3$, as constructed in Ammari et al. [5]. Then, there exist constants $\kappa \in \mathbb{R}$ and $R > 0$ such that for any $|r| > R$,*

$$c_r = \frac{\kappa}{|r|(\log|r|)^2} + o\left(\frac{1}{|r|(\log|r|)^2}\right)$$

*as $|r| \to \infty$.*

Theorem 2.4 establishes that the interaction coefficients decay as the separation between resonators increases. Although the decay is slower than $|r|^{-p}$ for any $p > 1$, the coefficients are nevertheless absolutely summable.

**Corollary 2.5** (Absolute summability and boundedness)**.** *Let $C = (c_{i-j})_{i,j\in\mathbb{Z}}$ be the infinite capacitance operator described in Theorem 2.4. Then*

$$\sum_{r\in\mathbb{Z}} |c_r| < \infty.$$

*Consequently, $C$ defines a bounded operator on $\ell^2(\mathbb{Z})$, satisfying*

$$\|C\|_{\mathrm{op}} \leq \sum_{r\in\mathbb{Z}} |c_r| < \infty.$$

*Proof.* By Theorem 2.4, there exist constants $K > 0$ and $R > 1$ such that

$$|c_r| \leq \frac{K}{|r|(\log|r|)^2}, \qquad |r| > R.$$

By the integral test,

$$\int_2^\infty \frac{1}{x(\log x)^2}\,dx = \frac{1}{\log 2} < \infty.$$

Hence,

$$\sum_{r\in\mathbb{Z}} |c_r| < \infty.$$

Since $C$ is a convolution operator with coefficients $(c_r)_{r\in\mathbb{Z}}$, Young's convolution inequality gives

$$\|C\psi\|_2 \leq \left(\sum_{r\in\mathbb{Z}} |c_r|\right) \|\psi\|_2$$

for all $\psi \in \ell^2(\mathbb{Z})$. Therefore,

$$\|C\|_{\mathrm{op}} \leq \sum_{r\in\mathbb{Z}} |c_r| < \infty.$$

□

Therefore, both $C$ and $H$ exhibit long-range interactions. Unlike the random band matrices introduced in Chapter 1, their off-diagonal entries are not restricted to a fixed bandwidth. However, the absolute summability of the coefficients suggests that interactions between sufficiently distant resonators may be discarded with a controllable approximation error.

This motivates the finite-band approximation introduced in the following subsection.

## 2.3 Finite-Band Approximation

After understanding a bit better of the mathematical model and its properties used in [5] to model acoustic resonators, we now consider a more general class of random operators satisfying similar decay properties, without requiring the coefficients to arise explicitly from an acoustic resonator geometry. We aim to determine how accurately these band operators approximate the original long-range operator as the bandwidth increases.

**Assumption 2.6** (Long-range random operator model)**.** Let $C = (c_{i-j})_{i,j\in\mathbb{Z}}$ be a Toeplitz operator with coefficients satisfying:

1. *(Real & Symmetric).* $c_{-r} = c_r \in \mathbb{R}$ for all $r \in \mathbb{Z}$.
2. *(Entries decay sufficiently quickly).*There exist constants $K > 0$ and $r_0 \geq 3$ such that

$$|c_r| \leq \frac{K}{|r|(\log|r|)^2} \qquad |r| \geq r_0$$

Note that this is obtained by the bound in Theorem 2.4.

Furthermore, let $(\gamma_i)_{i\in\mathbb{Z}}$ be positive random variables such that

$$0 < a \le \gamma_i \le b < \infty \qquad i \in \mathbb{Z}$$

almost surely, simultaneously for all $i$, where $a, b$ are deterministic constants.

We consider the random operator

$$H = V^{1/2} C V^{1/2} \qquad V = \operatorname{diag}(\gamma_i)_{i\in\mathbb{Z}}$$

**Definition 2.7** (Finite-band approximation)**.** Let $W \in \mathbb{N}$. Under Assumption 2.6, define the finite-band capacitance operator $C^{[W]}$ by

$$C^{[W]}_{ij} = \begin{cases} c_{i-j} & |i-j| < W \\ 0 & |i-j| \ge W \end{cases} \qquad i, j \in \mathbb{Z}$$

We define the corresponding random band operator by

$$H^{[W]} = V^{1/2} C^{[W]} V^{1/2}$$

Equivalently,

$$H^{[W]}_{ij} = \begin{cases} \sqrt{\gamma_i \gamma_j}\, c_{i-j} & |i-j| < W \\ 0 & |i-j| \ge W \end{cases}$$

Then, we claim the following proposition:

**Proposition 2.8** (Operator-norm convergence)**.** *Under Assumption 2.6, the finite-band approximations satisfy*

$$\|C - C^{[W]}\|_{\mathrm{op}} = O\left(\frac{1}{\log W}\right)$$

*as $W \to \infty$.*

*Furthermore, almost surely,*

$$\|H - H^{[W]}\|_{\mathrm{op}} = O\left(\frac{1}{\log W}\right).$$

*The implied constants may be chosen independently of the random realization.*

*Proof.* By Definition 2.7, we have

$$(C - C^{[W]})_{ij} = \begin{cases} 0, & |i-j| < W, \\ c_{i-j}, & |i-j| \ge W. \end{cases}$$

Thus, $C - C^{[W]}$ is a convolution operator whose coefficients are $c_r \mathbf{1}_{\{|r|\ge W\}}$.

By Young's convolution inequality,

$$\|C - C^{[W]}\|_{\mathrm{op}} \le \sum_{|r|\ge W} |c_r|.$$

Using Assumption 2.6, for all sufficiently large $W$,

$$\begin{aligned}\|C - C^{[W]}\|_{\mathrm{op}} &\le \sum_{|r|\ge W} |c_r| \\ &\le 2K \sum_{r=W}^{\infty} \frac{1}{r(\log r)^2}.\end{aligned}$$

By the integral test, for $W \ge 4$,

$$\begin{aligned}\sum_{r=W}^{\infty} \frac{1}{r(\log r)^2} &\le \int_{W-1}^{\infty} \frac{dx}{x(\log x)^2} \\ &= \frac{1}{\log(W-1)}.\end{aligned}$$

Consequently,

$$\|C - C^{[W]}\|_{\mathrm{op}} \le \frac{2K}{\log(W-1)} = O\left(\frac{1}{\log W}\right).$$

For the random operator, observe that

$$H - H^{[W]} = V^{1/2}(C - C^{[W]})V^{1/2}.$$

By submultiplicativity of the operator norm,

$$\begin{aligned}\|H - H^{[W]}\|_{\mathrm{op}} &\le \|V^{1/2}\|_{\mathrm{op}}^2 \|C - C^{[W]}\|_{\mathrm{op}} \\ &\le b\|C - C^{[W]}\|_{\mathrm{op}} \\ &= O\left(\frac{1}{\log W}\right).\end{aligned}$$

This holds almost surely, with constants independent of the random realization.

□

# 3 Fractional-Moment Analysis of the Resonator Model

As discussed in Section 1.3, Schenker [8] establishes localisation for certain random band matrices by studying two properties of their Green's functions:

1. Large values of the Green's function occur with small probability (Lemma W).
2. Fluctuations of the Green's function produce an improvement to Hölder's inequality that grows with the distance between sites (Lemma F).

In this section, we investigate whether these properties remain valid for the finite-band resonator operators introduced in Chapter 2. Our objective is not to establish a general localisation theorem, but rather to identify which arguments can be adapted to the resonator model and which require additional assumptions.

## 3.1 Random Material Parameters

Recall that the finite-band resonator operator is

$$H^{[W]} = V^{1/2} C^{[W]} V^{1/2} \qquad V = \operatorname{diag}(\gamma_i)$$

In the physical model, the capacitance coefficients describe interactions between the resonators in the resonator chain. Disorder may be introduced through variations in the material or build quality of individual resonators.

Motivated by this and by the random material perturbations studied in [5], we impose the following additional assumptions.

**Assumption 3.1** (Independent material disorder)**.** We suppose that:

1. The geometry of the resonator chain is fixed, so $C^{[W]}$ is deterministic.
2. The material parameters are given by

$$\gamma_i = \frac{1}{1 + \varepsilon \xi_i}$$

where

$$\xi_i \overset{\text{i.i.d.}}{\sim} \operatorname{Unif}[-1, 1] \qquad 0 < \varepsilon < 1$$

Here, the parameter $\varepsilon$ measures the strength of material disorder. In particular,

$$\frac{1}{1+\varepsilon} \leq \gamma_i \leq \frac{1}{1-\varepsilon}$$

Thus, Assumption 3.1 is consistent with the boundedness conditions in Assumption 2.6.

**Remark 3.2.** For the remainder of this section, we work with finite-volume restrictions to $N$ sites. We write these matrices as $H_N^{[W]}$ and $C_N^{[W]}$.

### 3.2 A Reformulation of the Green's Function

We take a look at the matrix $H_N^{[W]}$ and $C_N^{[W]}$ for say $W = 2$.

$$C_N^{[2]} = \begin{pmatrix} c_0 & c_1 & 0 & \cdots & 0 \\ c_1 & c_0 & c_1 & \ddots & \vdots \\ 0 & c_1 & c_0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & c_1 \\ 0 & \cdots & 0 & c_1 & c_0 \end{pmatrix} \qquad H_N^{[2]} = \begin{pmatrix} \gamma_1 c_0 & \sqrt{\gamma_1\gamma_2}c_1 & 0 & \cdots & 0 \\ \sqrt{\gamma_1\gamma_2}c_1 & \gamma_2 c_0 & \sqrt{\gamma_2\gamma_3}c_1 & \ddots & \vdots \\ 0 & \sqrt{\gamma_2\gamma_3}c_1 & \gamma_3 c_0 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \sqrt{\gamma_{N-1}\gamma_N}c_1 \\ 0 & \cdots & 0 & \sqrt{\gamma_{N-1}\gamma_N}c_1 & \gamma_N c_0 \end{pmatrix}$$

We reformulate its resolvent in terms of a matrix with independent diagonal disorder and deterministic off-diagonal interactions. This will allow us to apply spectral-averaging arguments analogous to that which was used in Schenker's Lemma W.

**Proposition 3.3** (Resolvent reformulation)**.** *Let Assumption 3.1 hold, and let*

$$H_N^{[W]} = V_N^{1/2} C_N^{[W]} V_N^{1/2},$$

*where* $V_N = \operatorname{diag}(\gamma_1, \ldots, \gamma_N)$*. For any* $\lambda \in \mathbb{R}$ *such that* $\lambda \notin \sigma(H_N^{[W]})$*, we have*

$$(H_N^{[W]} - \lambda I)^{-1} = V_N^{-1/2} (C_N^{[W]} - \lambda V_N^{-1})^{-1} V_N^{-1/2}$$

*Proof.* Since $\gamma_i > 0$ for all $i$, the diagonal matrix $V_N$ is positive definite and hence invertible.

By the definition of $H_N^{[W]}$, we have

$$\begin{aligned} H_N^{[W]} - \lambda I &= V_N^{1/2} C_N^{[W]} V_N^{1/2} - \lambda I \\ &= V_N^{1/2} C_N^{[W]} V_N^{1/2} - \lambda V_N^{1/2} V_N^{-1} V_N^{1/2} \\ &= V_N^{1/2} (C_N^{[W]} - \lambda V_N^{-1}) V_N^{1/2} \end{aligned}$$

Since $\lambda \notin \sigma(H_N^{[W]})$, the matrix $H_N^{[W]} - \lambda I$ is invertible. As $V_N^{1/2}$ is also invertible, $C_N^{[W]} - \lambda V_N^{-1}$ must be invertible.

Taking inverses on both sides and using $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$, we obtain

$$(H_N^{[W]} - \lambda I)^{-1} = V_N^{-1/2} (C_N^{[W]} - \lambda V_N^{-1})^{-1} V_N^{-1/2}$$

□

For fixed $\lambda \in \mathbb{R}$, define

$$A_{\lambda,N}^{[W]} = C_N^{[W]} - \lambda V_N^{-1}.$$

Its matrix entries are

$$(A_{\lambda,N}^{[W]})_{ij} = \begin{cases} c_0 - \lambda(1 + \varepsilon\xi_i), & i = j, \\ c_{i-j}, & 0 < |i-j| < W, \\ 0, & |i-j| \geq W. \end{cases}$$

Hence, $A_{E,N}^{[W]}$ is a random band matrix with independent diagonal disorder and deterministic off-diagonal interactions as we desired.

This is a useful connection with the Anderson model shown in Chapter 1.1. However, unlike an ordinary Anderson Hamiltonian, the distribution of the diagonal entries depends on the chosen spectral parameter $\lambda$.

Let

$$\widetilde{G}^{[W]}_{\lambda,N}(i,j) = \langle e_i, (A^{[W]}_{\lambda,N})^{-1} e_j \rangle.$$

The resolvent reformulation implies

$$G^{[W]}_N(i,j;\lambda) = \frac{\widetilde{G}^{[W]}_{\lambda,N}(i,j)}{\sqrt{\gamma_i \gamma_j}}.$$

Since $0 < \gamma_i^{-1} \leq 1 + \varepsilon$

$$|G^{[W]}_N(i,j;\lambda)| \leq (1+\varepsilon)|\widetilde{G}^{[W]}_{E,N}(i,j)|.$$

We will use this inequality to investigate Lemma W.

### 3.3 Investigation of Lemma W

We begin by studying the distribution of the random diagonal entries of $A^{[W]}_{\lambda,N}$.

For fixed $\lambda \neq 0$, define

$$q_i = -\lambda(1 + \varepsilon\xi_i).$$

Since $\xi_i$ is uniformly distributed on $[-1,1]$, it follows that $q_i \sim \mathrm{Unif}[-\lambda - \varepsilon|\lambda|, -\lambda + \varepsilon|\lambda|]$. In particular, their common density $\rho_\lambda$ satisfies

$$\|\rho_\lambda\|_\infty = \frac{1}{2\varepsilon|\lambda|}.$$

Thi will be useful for spectral averaging in Lemma $W$.

**Proposition 3.4** (Weak-tail estimate)**.** *Suppose Assumption 3.1 holds. Fix $\lambda \in \mathbb{R} \setminus \{0\}$.*

*Then the Green's function of $H^{[W]}_N$ exists almost surely at $\lambda$, and*

$$\mathbb{P}\left(|G^{[W]}_N(i,j;\lambda)| > t\right) \leq \frac{2(1+\varepsilon)}{\varepsilon|\lambda|t}$$

*for every $t > 0$ and all $1 \leq i, j \leq N$. Notice that the bound is independent of $N$ and $W$.*

*Proof.* Consider the transformed matrix

$$A = A^{[W]}_{\lambda,N} = C^{[W]}_N - \lambda V_N^{-1}.$$

Its random part consists of independent diagonal variables $q_i$ with bounded density $\rho_\lambda$, as shown. Fix $i \neq j$ and condition on all random variables except $q_i$ and $q_j$. By a change of basis, we may place the coordinates $i$ and $j$ first and write

$$A = \begin{pmatrix} D & T \\ T^* & R \end{pmatrix},$$

where $D$ is a $2 \times 2$ Hermitian matrix, $R$ is an $(N-2) \times (N-2)$ Hermitian matrix, and $T$ is a $2 \times (N-2)$ matrix. In particular, $D$ contains the random variables $q_i$ and $q_j$ only on its diagonal, while $T$ and $R$ are independent of these variables.

We first observe that $R$ is invertible almost surely. Indeed, $R$ has independent random diagonal entries with continuous distributions. Its determinant is a nonzero polynomial in these diagonal variables, since the product of the diagonal variables appears with coefficient one. By successively conditioning on these variables, the probability that this determinant vanishes is zero.

Therefore, we may apply the Schur complement formula. In particular, the upper-left $2 \times 2$ block of $A^{-1}$ is given by

$$(A^{-1})_{\{i,j\}\times\{i,j\}} = (D - TR^{-1}T^*)^{-1}$$

Define the Schur complement

$$B(q_i, q_j) := D - TR^{-1}T^*$$

Since $T$ and $R$ are independent of $q_i$ and $q_j$, and $TR^{-1}T^*$ is Hermitian, we may express $B$ as

$$B(q_i, q_j) = \begin{pmatrix} q_i - a & z \\ \overline{z} & q_j - b \end{pmatrix}$$

where $a, b \in \mathbb{R}$ and $z \in \mathbb{C}$ are independent of $q_i$ and $q_j$.

Thus, the $(i, j)$-th entry of $A^{-1}$ corresponds to the $(1, 2)$-th entry of $B^{-1}$. Hence,

$$\begin{aligned} |\widetilde{G}^{[W]}_{\lambda,N}(i, j)| &= |(A^{-1})_{ij}| \\ &= |(B^{-1})_{12}| \\ &= |\langle e_1, B^{-1}e_2\rangle| \\ &\leq \|B^{-1}\|_{\mathrm{op}} \end{aligned}$$

where the last inequality follows from the Cauchy–Schwarz inequality and the definition of the operator norm.

Next, since $B$ is Hermitian, its eigenvalues $\mu_1, \mu_2$ are real. By the spectral theorem,

$$\|B^{-1}\|_{\mathrm{op}} = \max\left\{\frac{1}{|\mu_1|}, \frac{1}{|\mu_2|}\right\}$$

Consequently, if $\|B^{-1}\|_{\mathrm{op}} > u$, then at least one eigenvalue of $B$ satisfies

$$|\mu_k| < \frac{1}{u}$$

So, $B$ has at least one eigenvalue in the interval

$$I_u = (-u^{-1}, u^{-1}).$$

Thus,

$$\left\{\|B^{-1}\|_{\mathrm{op}} > u\right\} = \{\sigma(B) \cap I_u \neq \varnothing\}$$

Since $q_i$ and $q_j$ are independent random variables with bounded density $\rho_\lambda$, and each enters $B$ as a rank-one diagonal perturbation, we may apply the standard rank-one spectral-averaging inequality to each coordinate. Thus,

$$\begin{aligned} \mathbb{P}(\|B^{-1}\|_{\mathrm{op}} > u) &\leq \mathbb{E}\left[\operatorname{tr} \mathbf{1}_{I_u}(B)\right] \\ &\leq 2\|\rho_\lambda\|_\infty |I_u| \\ &= \frac{4\|\rho_\lambda\|_\infty}{u} \end{aligned}$$

For $i = j$, the one-site Schur complement gives

$$\widetilde{G}^{[W]}_{\lambda,N}(i,i) = \frac{1}{q_i - a}$$

where $a \in \mathbb{R}$ is independent of $q_i$. So,

$$\begin{aligned}\mathbb{P}\left(|\widetilde{G}^{[W]}_{\lambda,N}(i,i)| > u\right) &= \mathbb{P}(|q_i - a| < u^{-1}) \\ &\leq \frac{2\|\rho_\lambda\|_\infty}{u} \\ &\leq \frac{4\|\rho_\lambda\|_\infty}{u}\end{aligned}$$

Hence, the same bound holds for $i = j$.

Recall that

$$\|\rho_\lambda\|_\infty = \frac{1}{2\varepsilon|\lambda|}$$

Therefore,

$$\mathbb{P}\left(|\widetilde{G}^{[W]}_{\lambda,N}(i,j)| > u\right) \leq \frac{2}{\varepsilon|\lambda|u}$$

Finally, by Proposition 3.3, we have

$$|G^{[W]}_N(i,j;\lambda)| \leq (1+\varepsilon)|\widetilde{G}^{[W]}_{\lambda,N}(i,j)|$$

It follows that

$$\begin{aligned}&\mathbb{P}\left(|G^{[W]}_N(i,j;\lambda)| > t\right) \\ &\quad\leq \mathbb{P}\left(|\widetilde{G}^{[W]}_{\lambda,N}(i,j)| > \frac{t}{1+\varepsilon}\right) \\ &\quad\leq \frac{2(1+\varepsilon)}{\varepsilon|\lambda|t}\end{aligned}$$

And we got the bound as desired. □

**Corollary 3.5** (Bounded fractional moments)**.** *Under the assumptions of Proposition 3.4, for every* $0 < s < 1$*,*

$$\mathbb{E}\left(|G^{[W]}_N(i,j;\lambda)|^s\right) \leq \frac{1}{1-s}\left(\frac{2(1+\varepsilon)}{\varepsilon|\lambda|}\right)^s.$$

*Proof.* Let

$$K_\lambda = \frac{2(1+\varepsilon)}{\varepsilon|\lambda|}.$$

For convenience, write $G = G^{[W]}_N(i,j;\lambda)$. Then, using the tail integral formula,

$$\mathbb{E}|G|^s = s\int_0^\infty t^{s-1}\mathbb{P}(|G| > t)\,dt$$

By Proposition 3.4,

$$\mathbb{P}(|G| > t) \leq \min\left\{1, \frac{K_\lambda}{t}\right\}$$

Splitting the integral at $K_\lambda$, we obtain

$$\begin{aligned}\mathbb{E}|G|^s &\le s\int_0^{K_\lambda} t^{s-1}\,dt + sK_\lambda\int_{K_\lambda}^{\infty} t^{s-2}\,dt\\ &= K_\lambda^s + \frac{sK_\lambda^s}{1-s}\\ &= \frac{K_\lambda^s}{1-s}\end{aligned}$$

Substituting the definition of $K_\lambda$ gives the desired estimate. □

**Remark 3.6.** The estimate is uniform in the matrix dimension $N$ and bandwidth $W$, but not in the disorder strength $\varepsilon$ or spectral parameter $\lambda$. In particular, the upper bound diverges as $\lambda \to 0$ or $\varepsilon \to 0$. Thus, the argument establishes an analogue of Schenker's Lemma W at each fixed nonzero spectral parameter, but does not provide a uniform estimate near $\lambda = 0$ or in the zero-disorder limit.

The significance of the weak-tail estimate is that the fractional moment of the Green function is bounded above by a term that is not dependent on $i, jN$ or $W$. This guarantees us uniformly control of the fractional moments for our resonator model.

### 3.4 Exponential decay of the Green's function under weak coupling

In Section 3.3, we showed that the fractional moments of the Green's function are bounded independently of $N$ and $W$. However, the bound obtained does not depend on $|i-j|$. Hence, it does not tell us whether the Green's function becomes smaller when the two sites are far apart.

We now investigate whether exponential decay can be established for $W \ge 2$. Rather than using the matrix-valued fluctuation argument in Schenker [8], we use a recursive estimate obtained by removing one site at a time. This is a version of the fractional-moment method of Aizenman and Molchanov [1]. The argument requires an additional assumption: the off-diagonal capacitance coefficients must be sufficiently small compared with the randomness in the material parameters.

Recall from Section 3.1 that

$$\gamma_i = \frac{1}{1+\varepsilon\xi_i}, \qquad \xi_i \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}[-1,1].$$

For convenience, write

$$M_N = V_N^{-1} = \mathrm{diag}(m_1,\dots,m_N), \qquad m_i = 1+\varepsilon\xi_i.$$

Thus, $m_i$ is uniformly distributed on $[1-\varepsilon, 1+\varepsilon]$, and

$$H_N^{[W]} = M_N^{-1/2} C_N^{[W]} M_N^{-1/2}.$$

We shall also consider complex spectral parameters $z = \lambda + \mathrm{i}\eta$ with $\eta > 0$. Define

$$A_N(z) = C_N^{[W]} - zM_N, \qquad \widetilde{G}_N(i,j;z) = (A_N(z)^{-1})_{ij}.$$

The factorisation in Proposition 3.3 remains valid for complex $z$:

$$(H_N^{[W]} - zI)^{-1} = M_N^{1/2} A_N(z)^{-1} M_N^{1/2}.$$

The use of $z$ will allow us to state an estimate that is uniform as $\eta$ approaches zero.

We first explain the additional assumption. Let $I \subset \mathbb{R}$ be a compact interval which does not contain zero, and define

$$\lambda_* := \inf_{\lambda \in I} |\lambda|.$$

Since $I$ is compact and $0 \notin I$, we have $\lambda_* > 0$. The quantity $\lambda_*$ is simply the smallest absolute value of the spectral parameter in the interval. For example, if $I = [2, 3]$ then $\lambda_* = 2$; if $I = [-3, -1]$ then $\lambda_* = 1$. We use $\lambda_*$ because a single constant will then work for every $\lambda \in I$.

Fix $0 < s < 1$. The random diagonal part of $A_N(z)$ is $-zm_i$. A useful calculation is that, for any $w \in \mathbb{C}$,

$$\begin{aligned}
\mathbb{E}_{m_i} |m_i - w|^{-s} &= \frac{1}{2\varepsilon} \int_{1-\varepsilon}^{1+\varepsilon} |x - w|^{-s} \, dx \\
&\leq \frac{1}{2\varepsilon} \int_{-\varepsilon}^{\varepsilon} |x|^{-s} \, dx \\
&= \frac{1}{(1-s)\varepsilon^s}
\end{aligned}$$

The inequality follows because $|x - w| \geq |x - \operatorname{Re} w|$, and the integral of $|x - a|^{-s}$ over an interval of fixed length is largest when the interval is centred at $a$. In particular, the integral is finite precisely because $s < 1$.

For $\operatorname{Re} z \in I$, we have $|z| \geq \lambda_*$. Hence,

$$\mathbb{E}_{m_i} |zm_i - w|^{-s} \leq \frac{1}{(1-s)(\varepsilon\lambda_*)^s}$$

We denote the constant on the right by

$$M_s := \frac{1}{(1-s)(\varepsilon\lambda_*)^s}$$

This constant measures how large the fractional moment of a single random diagonal denominator can be.

Next, notice that each site interacts with at most two other sites at each distance $r = 1, \ldots, W-1$. We therefore write

$$S_s := 2 \sum_{r=1}^{W-1} |c_r|^s, \qquad \theta_s := M_s S_s = \frac{2}{(1-s)(\varepsilon\lambda_*)^s} \sum_{r=1}^{W-1} |c_r|^s$$

The factor 2 accounts for sites to the left and right. The reason for introducing $\theta_s$ will become clear in the proof: it is an upper bound for the total weight gained at each step of a recursive Green's function estimate. If $\theta_s < 1$, repeated steps give a convergent geometric series. This motivates the following assumption.

**Proposition 3.7** (Exponential fractional-moment decay)**.** *Suppose Assumption 3.1 holds and fix a finite bandwidth $W \geq 2$. Let $I \subset \mathbb{R}$ be a compact interval with $\lambda_* > 0$. Suppose that for some $0 < s < 1$,*

$$\theta_s = \frac{2}{(1-s)(\varepsilon\lambda_*)^s} \sum_{r=1}^{W-1} |c_r|^s < 1$$

*Then there exist constants $C_s < \infty$ and $\mu_s > 0$, independent of $N, i, j$, such that*

$$\sup_{\substack{\lambda \in I \\ \eta > 0}} \mathbb{E}\left[ |G_N^{[W]}(i, j; \lambda + \mathrm{i}\eta)|^s \right] \leq C_s e^{-\mu_s |i-j|}$$

*The same estimate holds for each fixed real $\lambda \in I$, where the resolvent exists almost surely.*

*Proof.* We first work with the transformed Green's function $\widetilde{G}$. Let $\Lambda$ be any nonempty subset of $\{1, \dots, N\}$, and let $A_\Lambda(z)$ denote the principal submatrix of $A_N(z)$ with indices in $\Lambda$. We write

$$\widetilde{G}_\Lambda(i, j; z) = (A_\Lambda(z)^{-1})_{ij} \qquad i, j \in \Lambda$$

We introduce these smaller matrices because the proof will remove sites successively. Since $\operatorname{Im} z > 0$ and $M_N$ is positive, $A_\Lambda(z)$ is invertible for every choice of the material parameters.

Fix $i \in \Lambda$ and put the $i$-th coordinate first. We may write

$$A_\Lambda(z) = \begin{pmatrix} c_0 - zm_i & t_i^* \\ t_i & R_i(z) \end{pmatrix}$$

where $R_i(z) = A_{\Lambda\setminus\{i\}}(z)$ and the vector $t_i$ contains the capacitance coefficients connecting site $i$ to the remaining sites. Both $t_i$ and $R_i(z)$ are independent of $m_i$. By the Schur complement formula,

$$\widetilde{G}_\Lambda(i, i; z) = \big(c_0 - zm_i - t_i^* R_i(z)^{-1} t_i\big)^{-1}$$

The expression in parentheses is of the form $w - zm_i$, with $w$ independent of $m_i$. Averaging over $m_i$ and using the calculation above gives

$$\mathbb{E}_{m_i}\left[ |\widetilde{G}_\Lambda(i, i; z)|^s \,\middle|\, (m_k)_{k\neq i} \right] \le M_s$$

This bound holds for every subset $\Lambda$, regardless of how many sites have been removed.

We next consider $i \neq j$. The off-diagonal part of the same block inverse formula gives

$$\widetilde{G}_\Lambda(i, j; z) = -\widetilde{G}_\Lambda(i, i; z) \sum_{\substack{k\in\Lambda \\ k\neq i}} c_{i-k}^{[W]} \widetilde{G}_{\Lambda\setminus\{i\}}(k, j; z)$$

where $c_r^{[W]} = c_r$ for $|r| < W$ and is zero otherwise. This identity says that, to connect $i$ to $j$, we first move from $i$ to some other site $k$ and then use the Green's function of the matrix with site $i$ removed.

For $0 < s < 1$, we have $|\sum_k a_k|^s \le \sum_k |a_k|^s$. Applying this inequality, then averaging over $m_i$, gives

$$\mathbb{E}|\widetilde{G}_\Lambda(i, j; z)|^s \le M_s \sum_{\substack{k\in\Lambda \\ k\neq i}} |c_{i-k}^{[W]}|^s \mathbb{E}|\widetilde{G}_{\Lambda\setminus\{i\}}(k, j; z)|^s$$

The conditional expectation is possible because the resolvent on $\Lambda \setminus \{i\}$ no longer depends on $m_i$.

We now repeat this estimate. At each step, we remove the site from which we are moving, so a path cannot return to a site already removed. The iteration stops upon reaching $j$, when the remaining diagonal Green's function contributes at most $M_s$. For every site, the sum of the possible weights of the next step is at most

$$M_s \sum_{k\neq i} |c_{i-k}^{[W]}|^s \le 2M_s \sum_{r=1}^{W-1} |c_r|^s = \theta_s$$

Thus, the total contribution from paths of length $n$ is bounded by $M_s \theta_s^n$. There is also a restriction coming from the bandwidth. Since $c_{i-k}^{[W]} = 0$ for $|i - k| \ge W$, one step can move at most $W - 1$ sites. A path joining $i$ to $j$ must therefore have at least

$$n_0 = \left\lceil \frac{|i - j|}{W - 1} \right\rceil$$

steps. Summing over the possible path lengths, we obtain

$$\begin{aligned}\mathbb{E}|\widetilde{G}_N(i,j;z)|^s &\le M_s \sum_{n=n_0}^{\infty} \theta_s^n \\ &= \frac{M_s \theta_s^{n_0}}{1-\theta_s} \\ &\le \frac{M_s}{1-\theta_s} \exp\left(-\frac{|\log \theta_s|}{W-1}|i-j|\right)\end{aligned}$$

The geometric series converges because $\theta_s < 1$. The diagonal case $i = j$ follows from the one-site estimate and is covered by the same upper bound.

If $0 < \theta_s < 1$, set

$$\mu_s = \frac{|\log \theta_s|}{W-1} > 0$$

If $\theta_s = 0$, all off-diagonal couplings vanish, so the matrix is diagonal and the conclusion is immediate.

Finally, by the resolvent reformulation,

$$G_N^{[W]}(i,j;z) = \sqrt{m_i m_j}\, \widetilde{G}_N(i,j;z)$$

Since $m_i, m_j \le 1 + \varepsilon$, we have

$$|G_N^{[W]}(i,j;z)|^s \le (1+\varepsilon)^s |\widetilde{G}_N(i,j;z)|^s$$

The proposition follows, for example, with

$$C_s = \frac{(1+\varepsilon)^s M_s}{1-\theta_s}$$

For real $\lambda \in I$, almost-sure invertibility follows from the independent continuous diagonal variables, as in Section 3.3. Taking $\eta \downarrow 0$ and applying Fatou's lemma gives the same bound at the real spectral parameter. □

**Remark 3.8.** The additional condition $\theta_s < 1$ is a sufficient condition for exponential decay; it is not claimed to be necessary. It places a restriction on the relative sizes of the capacitance couplings and the random diagonal terms. For fixed $W$, the sum in $\theta_s$ contains only finitely many coefficients, so the condition can hold when the couplings are sufficiently weak.

However, our decay assumption from Section 2 only guarantees summability of $|c_r|$, not necessarily of $|c_r|^s$. Thus, the present argument does not give constants uniform in $W$ as $W \to \infty$. For $W = 2$, Schenker's tridiagonal theorem gives a result without this additional weak-coupling condition.

### 3.5 Verifying eigenvector localisation

We have shown that the fractional moments of the Green's function decay exponentially under the condition $\theta_s < 1$. This is an analogue of the Green's function estimate in Schenker's Theorem 1. However, our aim is to study the eigenvectors of $H_N^{[W]}$. As in Schenker's Theorem 2, we would like to show that the overlap of an eigenvector at two distant sites is small on average. This will be equivalent to showing eigenvector localisation for the finite-band approximation of the resonator model.

Let $\lambda_1, \dots, \lambda_N$ be the eigenvalues of $H_N^{[W]}$, with corresponding orthonormal eigenvectors $\psi_1, \dots, \psi_N$. For an interval $I$, consider the quantity

$$Q_I(i,j) := \sum_{\lambda_k \in I} |\psi_k(i)\psi_k(j)|$$

This sums the overlaps contributed by all eigenvectors whose eigenvalues lie in $I$. Bounding the sum is useful because

$$\sup_{\lambda_k \in I} |\psi_k(i)\psi_k(j)| \le Q_I(i,j)$$

We shall prove an exponential bound for $\mathbb{E}[Q_I(i,j)]$. The main additional step is to average over one material parameter while keeping all the other parameters fixed.

**Lemma 3.9** (One-site spectral averaging)**.** *Let $I$ be a compact interval with $\lambda_* > 0$. Fix a site $i$. For fixed values of $(m_\ell)_{\ell \neq i}$, let $f : I \to [0, \infty)$ be measurable, possibly depending on those fixed variables but not on $m_i$. Then*

$$\mathbb{E}_{m_i}\left[ \sum_{\lambda_k \in I} |\psi_k(i)|^2 f(\lambda_k) \,\middle|\, (m_\ell)_{\ell \neq i} \right] \le \frac{1+\varepsilon}{2\varepsilon\lambda_*} \int_I f(\lambda)\, d\lambda$$

*Proof.* We rewrite the eigenvalue equation using the diagonal matrix $M_N$. If $H_N^{[W]}\psi_k = \lambda_k \psi_k$, define

$$\phi_k = M_N^{-1/2}\psi_k$$

Substituting the definition of $H_N^{[W]}$ gives

$$C_N^{[W]}\phi_k = \lambda_k M_N \phi_k$$

Thus, the eigenvalue problem for $H_N^{[W]}$ can be written as a generalised eigenvalue problem involving the matrix $M_N$. Because $\psi_k$ is normalised,

$$\phi_k^* M_N \phi_k = 1$$

Keep $(m_\ell)_{\ell \neq i}$ fixed and vary only $m_i$. For a simple eigenvalue branch, differentiate $C_N^{[W]}\phi_k = \lambda_k M_N \phi_k$ with respect to $m_i$. Multiplying on the left by $\phi_k^*$ and using the eigenvalue equation cancels the terms involving the derivative of $\phi_k$. Since $dM_N/dm_i = e_i e_i^*$, we obtain

$$0 = \frac{d\lambda_k}{dm_i}\phi_k^* M_N \phi_k + \lambda_k |\phi_k(i)|^2$$

and therefore

$$\frac{d\lambda_k}{dm_i} = -\lambda_k |\phi_k(i)|^2 = -\frac{\lambda_k}{m_i}|\psi_k(i)|^2$$

For a branch with $\lambda_k \in I$, it follows that

$$|\psi_k(i)|^2\, dm_i = \frac{m_i}{|\lambda_k|}\,|d\lambda_k| \le \frac{1+\varepsilon}{\lambda_*}\,|d\lambda_k|$$

This explains why we require the interval to be away from zero: the factor $1/|\lambda_k|$ must be bounded.

We also need to see why integrating over different eigenvalue branches does not introduce a factor of $N$. To do this, delete the $i$-th row and column from $C_N^{[W]} - \lambda M_N$, and denote the

resulting matrix by $R_i(\lambda)$. For fixed $\lambda$, expansion of the determinant along its $i$-th diagonal entry shows that

$$\det(C_N^{[W]} - \lambda M_N)$$

is an affine function of $m_i$, with coefficient $-\lambda \det R_i(\lambda)$. Except at the finitely many values $\lambda = 0$ or $\det R_i(\lambda) = 0$, this coefficient is nonzero. Thus, for almost every $\lambda \in I$, there is at most one value of $m_i$ for which $\lambda$ is an eigenvalue of the full matrix.

We may therefore change variables from $m_i$ to $\lambda_k$ along the nonconstant eigenvalue branches, without counting the same energy more than once. Branches which are constant in $m_i$ have $\psi_k(i) = 0$ at nonzero energy and do not contribute. The same argument can be justified at isolated eigenvalue crossings by splitting the parameter interval into simple branches; these exceptional parameter values have zero probability under the continuous distribution of $m_i$.

Since $m_i$ has density $1/(2\varepsilon)$, the change of variables gives

$$\begin{aligned}
&\mathbb{E}_{m_i}\left[\sum_{\lambda_k \in I} |\psi_k(i)|^2 f(\lambda_k) \,\middle|\, (m_\ell)_{\ell \neq i}\right] \\
&\quad = \frac{1}{2\varepsilon}\int_{1-\varepsilon}^{1+\varepsilon} \sum_{\lambda_k \in I} |\psi_k(i)|^2 f(\lambda_k)\, dm_i \\
&\quad \leq \frac{1+\varepsilon}{2\varepsilon\lambda_*}\int_I f(\lambda)\, d\lambda
\end{aligned}$$

This proves the lemma. □

We now use this lemma to relate the Green's function estimate to the eigenvectors of the resonator matrix.

**Proposition 3.10** (Exponential decay of eigenvector overlaps)**.** *Under the assumptions of Proposition 3.7, there exist constants $C_s' < \infty$ and $\beta_s > 0$, independent of $N, i, j$, such that*

$$\mathbb{E}\left[\sum_{\lambda_k \in I} |\psi_k(i)\psi_k(j)|\right] \leq C_s' e^{-\beta_s |i-j|}$$

*In particular,*

$$\mathbb{E}\left[\sup_{\lambda_k \in I} |\psi_k(i)\psi_k(j)|\right] \leq C_s' e^{-\beta_s |i-j|}$$

*where the supremum over an empty set is taken to be zero. One may take $\beta_s = \mu_s/(2-s)$.*

*Proof.* Fix distinct sites $i, j$, and recall that $C_N^{[W]}\phi_k = \lambda_k M_N \phi_k$. We first express $\phi_k(j)$ in terms of $\phi_k(i)$. After deleting the $i$-th row and column, the equations at the remaining sites can be written as

$$R_i(\lambda_k)\phi_{k,\setminus i} = -t_i \phi_k(i)$$

where $R_i(\lambda) = C_{\setminus i}^{[W]} - \lambda M_{\setminus i}$ and $t_i$ contains the couplings between site $i$ and the other sites. Whenever $R_i(\lambda_k)$ is invertible,

$$\phi_{k,\setminus i} = -R_i(\lambda_k)^{-1} t_i \phi_k(i)$$

For $j \neq i$, this gives

$$\phi_k(j) = F_{j,i}(\lambda_k)\phi_k(i)$$

where we have set

$$F_{j,i}(\lambda) := -\sum_{\substack{\ell \neq i \\ |\ell - i| < W}} c_{\ell - i} \, [R_i(\lambda)^{-1}]_{j\ell}$$

The function $F_{j,i}(\lambda)$ does not depend on $m_i$. The exceptional energies at which $R_i(\lambda)$ is singular are finite in number. For almost every choice of $m_i$, any eigenvalue of the full matrix at one of these energies has zero amplitude at site $i$, so it does not contribute to $Q_I(i,j)$.

Since $\psi_k = M_N^{1/2} \phi_k$, we obtain

$$|\psi_k(j)| = \sqrt{\frac{m_j}{m_i}} \, |F_{j,i}(\lambda_k)| \, |\psi_k(i)|$$

In particular, the ratio between the two eigenvector amplitudes is controlled by a Green's function of the matrix with site $i$ removed.

We next estimate $F_{j,i}$. The fractional-moment argument in the preceding proposition was proved for every principal submatrix, so it applies to $R_i(\lambda)$. For fixed real $\lambda \in I$,

$$\begin{aligned} \mathbb{E}|F_{j,i}(\lambda)|^s &\leq \sum_{\substack{\ell \neq i \\ |\ell - i| < W}} |c_{\ell - i}|^s \mathbb{E}|[R_i(\lambda)^{-1}]_{j\ell}|^s \\ &\leq \frac{M_s}{1 - \theta_s} \sum_{\substack{\ell \neq i \\ |\ell - i| < W}} |c_{\ell - i}|^s e^{-\mu_s |j - \ell|} \end{aligned}$$

Every site $\ell$ appearing in the sum satisfies $|i - \ell| \leq W - 1$. By the triangle inequality, $|j - \ell| \geq |i - j| - (W - 1)$. Hence

$$\mathbb{E}|F_{j,i}(\lambda)|^s \leq \frac{M_s S_s e^{\mu_s (W-1)}}{1 - \theta_s} e^{-\mu_s |i-j|} =: D_s e^{-\mu_s |i-j|}$$

Thus $F_{j,i}$ also has an exponentially decaying fractional moment.

We now return to the eigenvector overlaps. For any $k$ with $\psi_k(i) \neq 0$, write

$$X_k = \frac{|\psi_k(j)|}{|\psi_k(i)|}.$$

Then

$$Q_I(i,j) = \sum_{\lambda_k \in I} |\psi_k(i)|^2 X_k.$$

We use Hölder's inequality to compare this expression with the $s$-th moment of $X_k$. Since

$$1 = \frac{s}{2 - s} + \frac{2(1 - s)}{2 - s},$$

Hölder's inequality, with exponents $2 - s$ and $(2 - s)/(1 - s)$, gives

$$Q_I(i,j) \leq \left( \sum_{\lambda_k \in I} |\psi_k(i)|^2 X_k^s \right)^{1/(2-s)} \times \left( \sum_{\lambda_k \in I} |\psi_k(i)|^2 X_k^2 \right)^{(1-s)/(2-s)}$$

But

$$\sum_{\lambda_k \in I} |\psi_k(i)|^2 X_k^2 \leq \sum_{k=1}^{N} |\psi_k(j)|^2 = 1$$

where the last equality follows from orthonormality of the eigenvectors. Therefore,

$$Q_I(i,j) \le \left( \sum_{\lambda_k \in I} |\psi_k(i)|^2 X_k^s \right)^{1/(2-s)}$$

The material parameters satisfy $1-\varepsilon \le m_i, m_j \le 1+\varepsilon$. Using the expression for $X_k$, we obtain

$$Q_I(i,j) \le \left(\frac{1+\varepsilon}{1-\varepsilon}\right)^{\frac{s}{2(2-s)}} \left( \sum_{\lambda_k \in I} |\psi_k(i)|^2 |F_{j,i}(\lambda_k)|^s \right)^{1/(2-s)}$$

Take expectations. Since $1/(2-s) < 1$, Jensen's inequality allows us to move the expectation inside the power. We then condition on all variables except $m_i$ and apply Lemma 3.9 to the function $f(\lambda) = |F_{j,i}(\lambda)|^s$, which is independent of $m_i$. This gives

$$\begin{aligned} \mathbb{E}Q_I(i,j) &\le \left(\frac{1+\varepsilon}{1-\varepsilon}\right)^{\frac{s}{2(2-s)}} \left( \frac{1+\varepsilon}{2\varepsilon\lambda_*} \int_I \mathbb{E}|F_{j,i}(\lambda)|^s \, d\lambda \right)^{1/(2-s)} \\ &\le \left(\frac{1+\varepsilon}{1-\varepsilon}\right)^{\frac{s}{2(2-s)}} \left( \frac{(1+\varepsilon)|I|D_s}{2\varepsilon\lambda_*} \right)^{1/(2-s)} e^{-\mu_s |i-j|/(2-s)} \end{aligned}$$

The constants depend on the fixed bandwidth and energy interval, but not on $N, i, j$. For $i = j$, we simply use $Q_I(i,i) \le \sum_k |\psi_k(i)|^2 = 1$. Thus, up to varying of constants, the estimate holds for all $i, j$, with $\beta_s = \mu_s/(2-s)$. Finally,

$$\sup_{\lambda_k \in I} |\psi_k(i)\psi_k(j)| \le \sum_{\lambda_k \in I} |\psi_k(i)\psi_k(j)|$$

which gives the second claim. □

**Remark 3.11.** We have therefore established an eigenvector localisation estimate for our finite-band resonator model for every fixed bandwidth $W \ge 2$, but under additional weak-coupling assumptions ($\theta_s < 1$). But unlike Schenker's result for Gaussian band matrices, our argument does not establish a localisation regime that allows the bandwidth $W$ to increase with $N$. So, the conclusion is uniform in $N$, but we have not shown that the constants remain uniform as $W$ increases. Consequently, this does not establish localisation for the original infinite-range resonator operator. It also does not show localisation at energies near zero, where our present estimates are not uniform.

### 3.6 Experiment 2: Visualising Localisation for the Resonator Model

We conclude with a numerical illustration of the localisation result obtained above. The code can be found in Appendix B.

For the simulation, we take

$$N = 50, \qquad W = 5, \qquad \varepsilon = 0.8, \qquad s = \frac{1}{2}, \qquad c_0 = 1$$

For the off-diagonal coefficients, we choose

$$c_r = \frac{0.02}{r(\log(r+1))^2}, \qquad r = 1, \dots, W-1.$$

Numerically, these coefficients are approximately

$$c_1 = 0.04163, \qquad c_2 = 0.00829, \qquad c_3 = 0.00347, \qquad c_4 = 0.00193.$$

Thus, $C_N^{[W]}$ is a symmetric band matrix of bandwidth $W = 5$. A single realisation of the random material parameters $\gamma_i$ is then generated, and the corresponding matrix $H_N^{[W]}$ is constructed as described.

We consider the energy interval $I = [4, 8]$ so that $\lambda_* = \inf_{\lambda \in I} |\lambda| = 4$. For these parameters, the sufficient condition appearing in the fractional-moment estimate gives

$$\theta_s = \frac{2}{(1-s)(\varepsilon\lambda_*)^s} \sum_{r=1}^{W-1} |c_r|^s \approx 0.8897 < 1$$

Hence, since $\theta_s < 1$, we expect by our result that the matrix is eigenvector-localised.

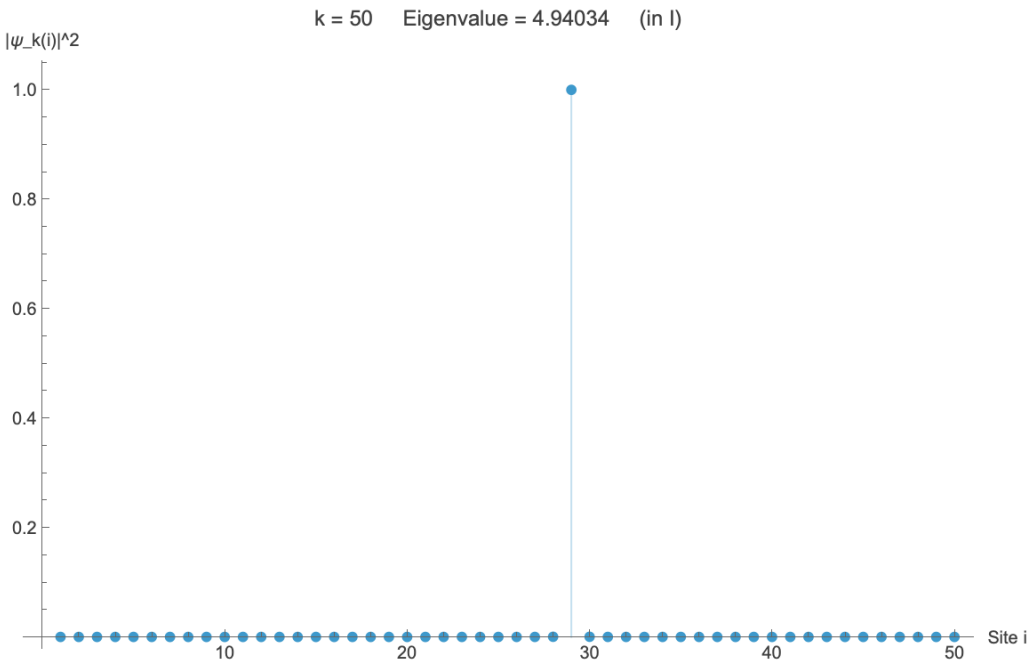



Figure 2: Spatial profile of a localised eigenvector of the random resonator matrix with $N = 50$, $W = 5$, and $\varepsilon = 0.8$, corresponding to an eigenvalue $\lambda \approx 4.94034$ within the interval $I = [4, 8]$.

From the numerical simulations, we see that the eigenvectors are strongly localised. For example, Figure 2 shows a representative eigenvector whose mass is concentrated near a single region of the resonator chain. Furthermore, this eigenvector is in the interval $I = [4, 8]$ and hence should (by our findings) be localised.

But one important remark is that in our weak-coupling regime, the off-diagonals are extremely small ($\sim 0.04$), and hence, it is very close to a diagonal matrix.

This raises the further question of whether localisation persists when the off-diagonal interactions are increased beyond the regime covered by our sufficient condition. We investigate this numerically in the following experiment.

To investigate this numerically, we keep $N = 50$, $W = 5$, $\varepsilon = 0.8$, $c_0 = 1$, and the same realisation of the random material parameters, while varying the interaction strength $a$ in

$$c_r(a) = \frac{a}{r(\log(r+1))^2} \qquad 1 \leq r < W.$$

For each value of $a$, we construct the corresponding random resonator matrix and compute the mean inverse participation ratio of the eigenvectors under consideration. Recall that

$$\sum_{i=1}^{N} |\psi(i)|^4$$

where a larger IPR corresponds to stronger spatial concentration. For example, a completely extended eigenvector with approximately equal mass on all $N = 50$ sites has

$$\text{IPR}(\psi) \approx \frac{1}{N} = 0.02$$

As the interaction strength increases, the mean IPR decreases, indicating that the eigenvectors become progressively less localised.

We plot the mean IPR as $\theta_s$ increases. Notice in Figure 3 that even when $\theta_s > 1$, the mean IPR still stays relatively high. This suggests that perhaps our choice of retriction of $\theta_s < 1$ is conservative, and there should be some bound that is perhaps around $\theta_s = 2$ where the eigenvectors start being less localised. And an inverse relationship is certainly expected.

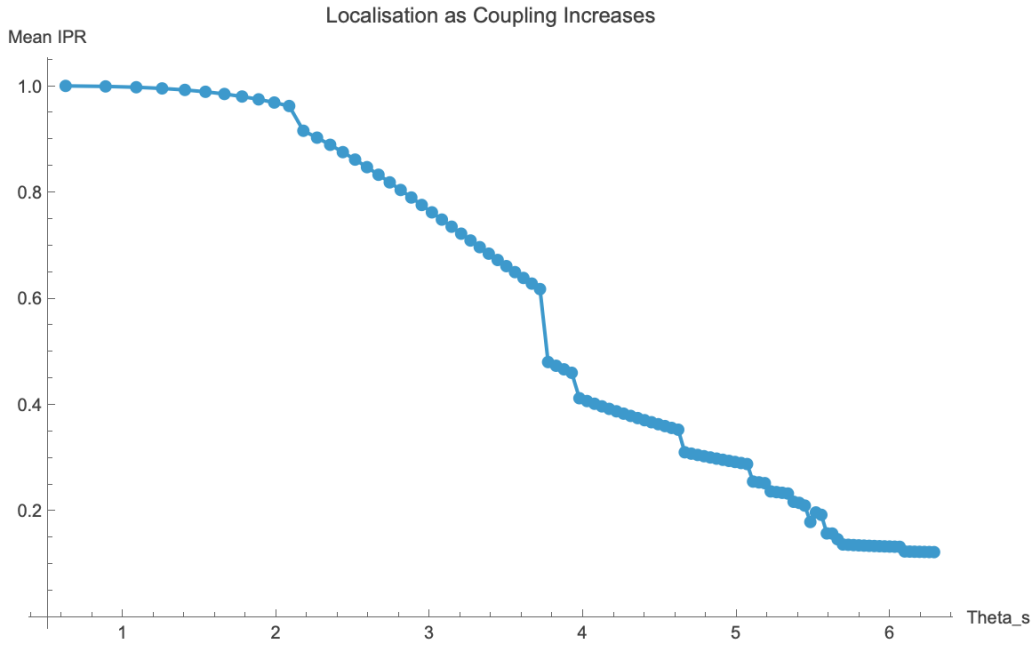



Figure 3: Mean inverse participation ratio as the off-diagonal interaction strength is increased.

# 4 Concluding Remarks

Our analysis in Chapter 3 shows that translating models to a finite-band regime can be interesting, and if given more restrictive assumptions, one may also show localisation for band matrices that model certain real-life scenarios.

In our case, the study of a finite-band acoustic resonator model allows us to study more in-depth the phenomenon of Anderson localisation in acoustic setups. More specifically, if studied more in-depth, we could perhaps get a bound for which Anderson localisation occurs. And this may have important implications in applied physics.

Establishing Anderson localisation for the full long-range resonator model remains an open problem, as also discussed in [5]. A natural direction for future research would be to investigate whether the weak-coupling assumption can be relaxed as shown in our numerical analysis, and whether localisation estimates can be obtained uniformly as the bandwidth increases. Such results would provide a stronger mathematical understanding of how disorder and long-range interactions influence the spatial distribution of acoustic resonances.

**Declaration of AI Use.** ChatGPT-6 was used during the preparation and development of this report, primarily to assist in refining mathematical arguments, reviewing proofs for potential errors, and developing and verifying numerical code.

# A Experiment 1 Code

Numerical experiment on localisation of Gaussian band matrices.

Code is in *Mathematica*.

```
SeedRandom[42];


n=500;
W=5 ;
h=ConstantArray[0.+0. I,{n,n}];

(*random diagonal entries*)
Do[h[[i,i]]=RandomVariate[NormalDistribution[0,1/Sqrt[4 W]]],{i,1,n}];

(*random complex off-diagonal entries*)
Do[If[Abs[i-j]<W,h[[i,j]]=RandomVariate[NormalDistribution[0,1/Sqrt[8 W]]]+I RandomVariate[NormalDistribution[0,1/Sqrt[8 W]]];
h[[j,i]]=Conjugate[h[[i,j]]];],{i,1,n},{j,i+1,n}];

(*print matrix*)
Print["Gaussian Random Band Matrix H:"];
Print[MatrixForm[Map[NumberForm[#,{5,3}]&,h,{2}]]];


{vals,vecs}=Eigensystem[h];
Manipulate[ListPlot[Abs[vecs[[k]]]^2,Filling->Axis,PlotRange->All,AxesLabel->{"Site i","|\[Psi]_k(i)|^2"},PlotLabel->Row[{"Eigenvalue = ",N[
      vals[[k]],5]}],ImageSize->Large],{k,1,n,1}]
```

# B Experiment 2 Code

Numerical construction of the random resonator matrix and computation of its eigenvectors and localisation parameter $\theta_s$.

Code is in *Mathematica*.

```
SeedRandom[42];

n = 50;
W = 5;
epsilon = 0.8;
s = 0.5;
c0 = 1;

(*lambda - interval where the eignvectors experience localisation*)
lambdaMin = 4;
lambdaMax = 8.0;
lambdaStar = 4.0;


Print["Matrix C:"];
couplingMax = 0.02;
couplings = Table[couplingMax/(r Log[r + 1]^2), {r, 1, W - 1}];
c = ConstantArray[0., {n, n}];
Do[c[[i, i]] = c0, {i, 1, n}];
Do[If[Abs[i - j] < W, c[[i, j]] = couplings[[Abs[i - j]]];
   c[[j, i]] = c[[i, j]];], {i, 1, n}, {j, i + 1, n}];
Print[MatrixForm[Map[NumberForm[#, {5, 3}] &, c, {2}]]];


xi = RandomReal[{-1, 1}, n];
gamma = 1/(1 + epsilon xi);
v = DiagonalMatrix[gamma];
vHalf = DiagonalMatrix[Sqrt[gamma]];


h = vHalf . c . vHalf;

Print["Resonator Matrix H:"];
Print[MatrixForm[Map[NumberForm[#, {5, 3}] &, h, {2}]]];




thetaS = 2 Total[Abs[couplings]^s]/((1 - s) (epsilon lambdaStar)^s);
Print["Theta_s = ", N[thetaS, 5]];
If[thetaS < 1, Print["Theta_s < 1: theorem condition satisfied"],
  Print["Theta_s >= 1: theorem condition NOT satisfied"]];




{vals, vecs} = Eigensystem[h];
order = Ordering[vals];
vals = vals[[order]];
vecs = vecs[[order]];



Manipulate[
 ListPlot[Abs[vecs[[k]]]^2, Filling -> Axis, PlotRange -> All,
  AxesLabel -> {"Site i", "|\[Psi]_k(i)|^2"},
  PlotLabel ->
   Row[{"k = ", k, "     Eigenvalue = ", N[vals[[k]], 5],
     If[lambdaMin <= vals[[k]] <= lambdaMax, "     (in I)",
      "     (outside I)"]}], ImageSize -> Large], {k, 1, n, 1}]




(*Experient 2 - Testing larger values*)
aValues = Range[0.01, 1.0, 0.01];

results = Table[
   (*New coupling coefficients*)
   couplingsA = Table[a/(r Log[r + 1]^2), {r, 1, W - 1}];
   (*New capacitance matrix*)
   cA = Table[
     If[i == j, c0,
      If[0 < Abs[i - j] < W, couplingsA[[Abs[i - j]]], 0]], {i,
      n}, {j, n}];
   (*Same disorder V,but stronger coupling*)  hA = vHalf . cA . vHalf;
   (*Theta_s*)
   thetaA =
    2 Total[Abs[couplingsA]^s]/((1 - s) (epsilon lambdaStar)^s);
   (*Eigenvectors*)  {valsA, vecsA} = Eigensystem[hA];
   (*IPR of each eigenvector*)  iprA = Total[Abs[#]^4] & /@ vecsA;
   (*Keep eigenvectors with eigenvalues in I*)
   indicesA = Select[Range[n], lambdaMin <= valsA[[#]] <= lambdaMax &];
   (*Return {coupling,theta_s,
   mean IPR}*)   {a, thetaA,
    If[Length[indicesA] > 0, Mean[iprA[[indicesA]]], Missing[]]}, {a,
    aValues}];
```

```
88
89
90
91 Print[Grid[Prepend[results, {"Coupling a", "Theta_s", "Mean IPR"}],
92    Frame -> All]];
93
94
95 validResults = Select[results, NumberQ[#[[3]]] &];
96 ListLinePlot[validResults[[All, {2, 3}]], PlotMarkers -> Automatic,
97  PlotRange -> All, AxesLabel -> {"Theta_s", "Mean IPR"},
98  PlotLabel -> "Localisation as Coupling Increases",
99  ImageSize -> Large]
```

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